

- 1 The European hedgehog, *Erinaceus europaeus*, is a small omnivorous mammal. Its body is covered in spines, which are usually brown. A rare variant, which lacks the brown pigment, has blonde (pale yellow) spines. This characteristic is coded for by a recessive allele.

Fig. 1.1 shows a European hedgehog with blonde spines.



Fig. 1.1

- (a) Alderney is a small island between the UK and France. Hedgehogs were **not** found on Alderney until the 1960s, when three pairs of hedgehogs were introduced to the island. The hedgehogs started to breed and some of the offspring had blonde spines.

By 2017, the population of hedgehogs, including individuals with brown spines and individuals with blonde spines, had increased to approximately 600.

- (i) Suggest reasons why the population of hedgehogs increased to such a large number.

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- (ii) A survey taken in 2017 showed that 60% of the hedgehogs on Alderney had blonde spines.

Suggest **and** explain reasons why the proportion of hedgehogs with blonde spines was so high.

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- (b) The hedgehogs on Alderney are an example of an invasive alien species.

Explain why it is sometimes important to control invasive alien species.

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